

# # VPC Gateway Endpoint

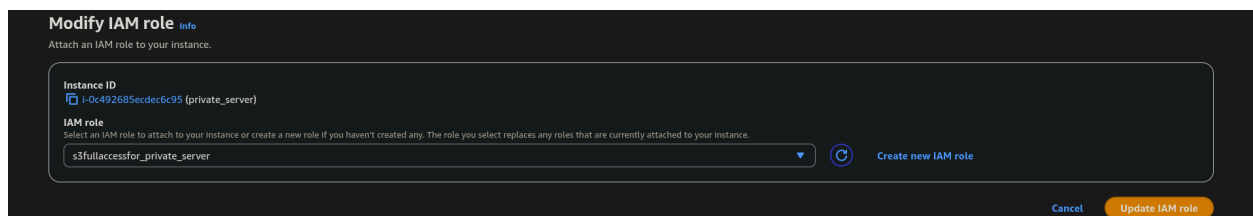
## #objective:

The objective of this lab is to understand how a **Gateway VPC Endpoint** works and how it allows private communication between resources inside a VPC and supported AWS services (Amazon S3 and DynamoDB) without using an Internet Gateway, NAT Gateway, or public internet.

## #Lab Environment

- 1 Amazon VPC
- 1 Private Subnet
- 1 Public Subnet
- 1 Amazon EC2 Instance
- Amazon S3 Bucket
- Route Tables
- Gateway VPC Endpoint

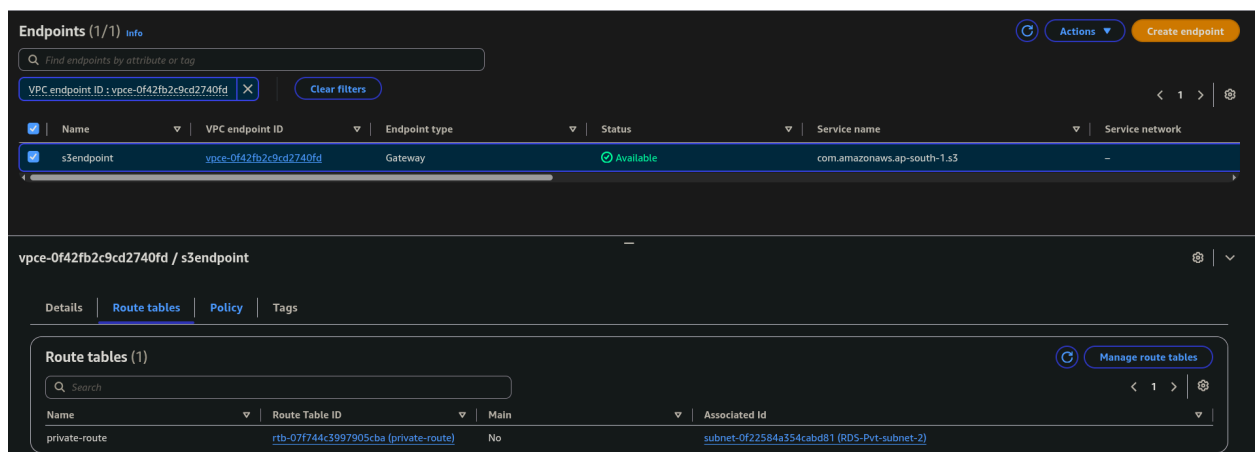
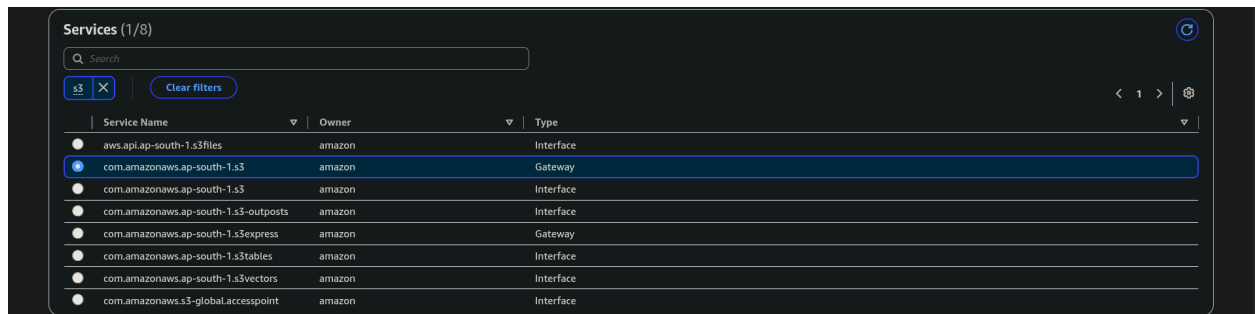
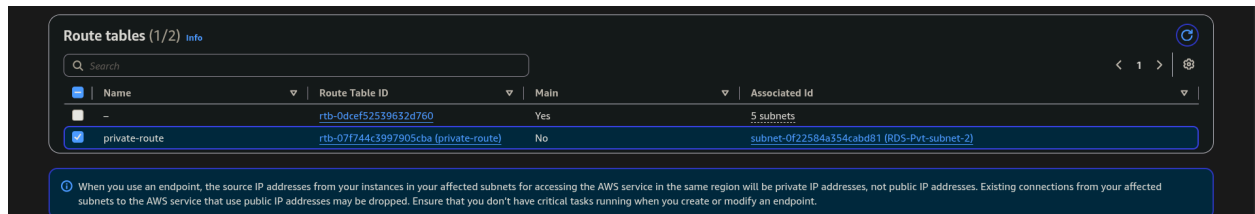
# First I launch 2 instances: one in private subnet and another in public subnet. Then I SSH into the private instance and try to list the S3 bucket, but I can't even attach the `s3FullAccess` role to this private instance.



```
/m/
[ec2-user@ip-172-31-48-194 ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2001ms

[ec2-user@ip-172-31-48-194 ~]$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-48-194 ~]$ aws s3 ls
```

# Create VPC Endpoint Gateway for give securely access s3 service , in creation mode there select VPC, route table and service



# After these configuration i try to again list s3 bucket then finally i can list then s3 bucket privately without going through internet



#Conclusion:

In this lab, I successfully created an **AWS Gateway VPC Endpoint** for Amazon S3. I verified that an EC2 instance located in a private subnet could securely access an S3 bucket without requiring an Internet Gateway or NAT Gateway. This exercise demonstrated how

Gateway VPC Endpoints provide secure, private, and cost-effective access to supported AWS services while ensuring that traffic remains within the AWS network.